

We prove that P is the midpoint of HQ and then compute the ratio.

1. Coordinates. Place $B = (0, 0)$, $C = (17, 0)$ and $A = (10, 24)$; this satisfies $AB = 26$, $AC = 25$. The triangle is acute (e.g. $17^2 + 25^2 > 26^2$).

2. Orthocenter and feet. BC horizontal altitude from A is $x = 10$. Slope of $AC = \frac{0-24}{17-10} = -\frac{24}{7}$ altitude from B : $y = \frac{7}{24}x$. Intersection gives $H = (10, \frac{35}{12})$; the foot $D = (10, 0)$.

Foot E from B to AC : line AC : $24x + 7y - 408 = 0$ $E = (\frac{24 \cdot 408}{625}, \frac{7 \cdot 408}{625}) = (\frac{9792}{625}, \frac{2856}{625})$.

Foot F from C to AB : line AB : $12x - 5y = 0$ $F = (\frac{425}{169}, \frac{1020}{169})$.

3. Point $X = EF \cap BC$. BC : $y = 0$. Using $x_X = \frac{x_E y_F - x_F y_E}{y_F - y_E}$ and noting common factor 17 in E, F we obtain $X = (\frac{170}{3}, 0)$.

4. Point Q – second intersection of AX with circumcircle. For the circumcircle, the power of X gives $XA \cdot XQ = XB \cdot XC$ (directed lengths). $XB = \frac{170}{3}$, $XC = \frac{119}{3} \Rightarrow XB \cdot XC = \frac{20230}{9}$. $XA = \sqrt{(\frac{140}{3})^2 + 24^2} = \frac{4\sqrt{1549}}{3}$ $XQ = \frac{XB \cdot XC}{XA} = \frac{20230}{12\sqrt{1549}}$; clearly $XQ < XA$ so Q lies between X and A .

Set $\mu = \frac{XQ}{XA} = \frac{XB \cdot XC}{XA^2} = \frac{20230}{24784} = \frac{10115}{12392}$. Then $Q = X + \mu(A - X)$. Writing $d = 1549$ and $K = 5 \cdot 7 \cdot 11 \cdot 23 = 8855$, simplification yields

$$Q - H = \left(\frac{3K}{2d}, \frac{35K}{12d} \right).$$

Hence

$$HQ^2 = \left(\frac{3K}{2d} \right)^2 + \left(\frac{35K}{12d} \right)^2 = K^2 \left(\frac{9}{4d^2} + \frac{1225}{144d^2} \right) = \frac{K^2 \cdot 1549}{144d^2} = \frac{K^2}{144d},$$

$$\text{so } HQ = \frac{K}{12\sqrt{d}}.$$

5. Identification of P . The nine-point circle (through D, E, F) is the image of the circumcircle under the homothety with centre H and factor $\frac{1}{2}$ (the midpoints of HA, HB, HC lie on it and its centre is the midpoint of OH , radius $R/2$). Since Q lies on the circumcircle, its image under this homothety – the midpoint of HQ – lies on the nine-point circle and on segment HQ (because H is inside the circumcircle for an acute triangle). Therefore the intersection P of segment HQ with the nine-point circle is precisely the midpoint of HQ . Consequently

$$PQ = \frac{1}{2}HQ = \frac{K}{24\sqrt{d}}.$$

6. Computation of PE . With $P = (10 + \frac{3K}{4d}, \frac{35}{12} + \frac{35K}{24d})$ and $E = (\frac{9792}{625}, \frac{2856}{625})$ write

$$U = E_x - P_x = \frac{9792}{625} - 10 - \frac{3K}{4d} = \frac{3542}{625} - \frac{3K}{4d},$$

$$V = E_y - P_y = \frac{2856}{625} - \frac{35}{12} - \frac{35K}{24d} = \left(\frac{2856}{625} - \frac{35}{12} \right) - \frac{35K}{24d}.$$

Using $3542 = 2 \cdot 1771$ and $1771 = K/5$ gives $3542 = 2K/5$. Also $\frac{2856}{625} - \frac{35}{12} = \frac{12397}{7500}$ and $12397 = 7 \cdot 1771 = 7K/5$.

Hence

$$U = K \left(\frac{2}{3125} - \frac{3}{4d} \right), \quad V = K \left(\frac{7}{37500} - \frac{35}{24d} \right).$$

Insert $d = 1549$:

$$8d - 9375 = 3017 = 7 \cdot 431, \quad 14d - 109375 = -87689 = -7 \cdot 12527,$$

so

$$U = K \cdot \frac{8d - 9375}{12500d} = \frac{7K \cdot 431}{12500d}, \quad V = K \cdot \frac{14d - 109375}{75000d} = -\frac{7K \cdot 12527}{75000d} = -\frac{7K \cdot 12527}{6 \cdot 12500d}.$$

Therefore

$$PE^2 = U^2 + V^2 = \frac{49K^2}{12500^2 d^2} \left(431^2 + \frac{12527^2}{36} \right) = \frac{49K^2}{12500^2 d^2} \cdot \frac{36 \cdot 431^2 + 12527^2}{36}.$$

Now $36 \cdot 431^2 + 12527^2 = 36 \cdot 185761 + 156925729 = 163613125 = 325^2 \cdot 1549 = 325^2 d$. Substituting,

$$PE^2 = \frac{49K^2}{12500^2 d^2} \cdot \frac{325^2 d}{36} = \frac{49K^2 \cdot 325^2}{12500^2 \cdot 36 d}.$$

Since $\frac{325}{12500} = \frac{13}{500}$, $\frac{325^2}{12500^2} = \frac{169}{250000}$. Thus

$$PE^2 = \frac{49K^2 \cdot 169}{250000 \cdot 36 d} = \frac{49K^2 \cdot 169}{9\,000\,000 d}, \quad \text{so} \quad PE = \frac{7 \cdot 13 \cdot K}{3000\sqrt{d}} = \frac{91K}{3000\sqrt{d}}.$$

7. Ratio. From (3) and (4),

$$\frac{PQ}{PE} = \frac{K/(24\sqrt{d})}{91K/(3000\sqrt{d})} = \frac{3000}{24 \cdot 91} = \frac{125}{91}.$$

Thus $a = 125$, $b = 91$ are coprime, and $a + b = 216$.

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